

Blockchain Final.DeFi-SuperApp-Capstone

Final Project Report

GitHub Repository

<https://github.com/sultansmailov3/BlockchainFinal.DeFi-SuperApp-Capstone>

Team Members

Sultan Kalzhigit,Zhantore Gaineden,Batyrkhan Kairzhanov

Participant	Role
Sultan	Smart Contract Lead
Zhantore	Security & DevOps Engineer
Batyrkhan	Full-Stack & Integration Lead

1. Introduction

The purpose of this project was to develop a decentralized finance (DeFi) SuperApp using Solidity smart contracts, governance systems, Web3 frontend technologies, and blockchain infrastructure tools.

The application combines automated token exchange, vault systems, governance voting, oracle integration, and upgradeable smart contracts into one decentralized ecosystem.

2. Project Objectives

Main objectives of the project:

Build a decentralized finance application

Implement AMM swap functionality

Create ERC-4626 vault contracts

Configure governance using OpenZeppelin

Integrate Chainlink oracles

Develop React frontend application

Implement secure upgradeability

Create automated testing infrastructure

3. Technology Stack

Technology	Purpose
Solidity	Smart contracts
Foundry	Testing
React	Frontend
Wagmi/Viem	Wallet interaction
OpenZeppelin	Security and governance
Chainlink	Oracle integration
The Graph	Blockchain indexing
GitHub Actions	CI/CD

4. Project Structure

BlockchainFinal.DeFi-SuperApp-Capstone/

- >  .github/workflows
- >  cache
- >  lib
- >  script
- >  src
- >  subgraph
- >  test
-  .gitignore
-  .gitmodules
-  App.css
-  LICENSE
-  README.md

The repository is separated into smart contracts, testing infrastructure, deployment scripts, frontend application, and indexing services.

5. Smart Contract Development

The smart contract layer was developed using Solidity.

Main components include:

AMM liquidity pool

ERC-4626 vault

Governance contracts

Factory deployment contract

Oracle integration contracts

AMM Logic

The AMM uses the constant product formula:

$$x \cdot y = k$$

A 0.3% swap fee was implemented together with slippage protection.

ERC-4626 Vault

The vault supports:

Deposit

Withdraw

Mint shares

Redeem shares

Security Features

Implemented protections:

Reentrancy Guard

Checks-Effects-Interactions pattern

Input validation

Oracle freshness checks

6.Security & DevOps

The Security & DevOps module focused on governance, testing, CI/CD automation, and deployment.

Governance

The governance stack was built using:

Governor

Timelock Controller

ERC20Votes

The system supports:

Proposal creation

Voting

Timelock execution

Testing

More than 80 tests were written using Foundry.

Testing included:

Unit tests

Invariant tests

Fork tests

CI/CD

GitHub Actions was configured to automatically run:

forge test

Slither analysis

Build checks

7. Frontend & Integration

The frontend was developed using React, Wagmi, and Viem.

Frontend features include:

Wallet connection

Token swaps

Vault interaction

Governance interaction

Error handling

The Graph

A subgraph was created for indexing blockchain events.

Indexed entities include:

Users

Pools

Swaps

Vault deposits

Chainlink Integration

Chainlink Price Feeds were integrated for secure asset pricing.

Oracle staleness checks were implemented to prevent outdated price usage.

8. Upgradeability

The project uses the UUPS proxy pattern.

Benefits:

Future upgrades

Lower deployment costs

Storage persistence

The upgrade process supports migration from V1 to V2 implementations.

9. Testing Results

Category	Result
Unit Tests	Passed
Invariant Tests	Passed
Fork Tests	Passed
Governance Tests	Passed
Slither Analysis	Completed
CI/CD Pipeline	Functional

10. Deployment

The project was deployed on:

Arbitrum Sepolia

Optimism Sepolia

Deployment scripts support:

Automatic verification

Multi-network deployment

Upgradeable contracts

11. Conclusion

The BlockchainFinal.DeFi-SuperApp-Capstone project successfully demonstrates a complete decentralized finance ecosystem using modern Ethereum development standards.

The project combines:

Solidity smart contracts

Governance infrastructure

Oracle integration

Upgradeability

React frontend development

Blockchain indexing

Security testing

Automated CI/CD pipelines

The project provided practical experience in decentralized application development, smart contract security, Web3 integration, and blockchain infrastructure engineering.